

Corollary 2.3.26 (Markov's inequality).

$$P(|X| \geq a) \leq \frac{1}{a^k} E(|X|^k).$$

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$$\text{var}(X) = E((X - \mu)^2) = \sum_x (x - \mu)^2 P(X = x).$$

The *standard deviation* $\sigma(X)$ of X is defined as the square root of the variance,

$$\sigma(X) = \sqrt{\text{var}(X)}.$$

Lemma 2.3.18. $\text{var}(X) = E(X^2) - (E(X))^2$.

♠ **Exercise 2.3.13.** Find two random variables X and Y so that

$$E(XY) \neq E(X)E(Y).$$

Theorem 2.3.14. If the random variables X and Y are independent and $E(X)$ and $E(Y)$ are finite, then $E(XY)$ is well defined and satisfies

$$E(XY) = E(X)E(Y).$$

♠ **Exercise 2.3.8.** Show that if X takes only non-negative integer values, we have

$$E(X) = \sum_{n=0}^{\infty} P(X > n).$$

Theorem 2.3.20. Let X and Y be random variables, and $a, b \in \mathbb{R}$.

- (a) $\text{var}(aX + b) = a^2 \text{var}(X)$.
 (b) $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y) + 2(E(XY) - E(X)E(Y))$. In particular, if X and Y are independent, then

$$\text{var}(X + Y) = \text{var}(X) + \text{var}(Y).$$

$$\begin{aligned} \text{var}(X + Y) &= E((X + Y - E(X + Y))^2) \\ &= E((X - E(X))^2 + (Y - E(Y))^2 + 2(XY - E(X)E(Y))) \\ &= \text{var}(X) + \text{var}(Y) + 2(E(XY) - E(X)E(Y)). \end{aligned}$$

□

Definition 2.3.21. The quantity $E(XY) - E(X)E(Y)$ which appears in (b) is called the *covariance* of X and Y , and denoted by $\text{cov}(X, Y)$.

♠ **Exercise 2.3.22.** Usually the covariance of X and Y is defined as $\text{cov}(X, Y) = E((X - E(X))(Y - E(Y)))$. Show that this amounts to the same definition as above.

Theorem 2.3.24. Let X be a random variable taking only non-negative values. Then for any $a > 0$ we have

$$P(X \geq a) \leq \frac{1}{a} E(X).$$

Proof.

$$\begin{aligned} E(X) &= \sum_x xP(X = x) \\ &\geq \sum_{x: x \geq a} xP(X = x) \geq a \sum_{x: x \geq a} P(X = x) \\ &= aP(X \geq a). \end{aligned}$$

□

- (a) $\lim_{x \rightarrow \infty} F(x) = 1$;
 (b) $\lim_{x \rightarrow -\infty} F(x) = 0$;
 (c) F is non-decreasing;
 (d) F is continuous from the right;
 (e) $P(X > x) = 1 - F(x)$;
 (f) $P(x < X \leq y) = F(y) - F(x)$;
 (g) $P(X = x) = F(x) - \lim_{y \uparrow x} F(y)$.

Markov's inequality for $k = 2$ and applied to $|X - E(X)|$ leads to

Corollary 2.3.27 (Chebyshev's inequality).

$$P(|X - E(X)| \geq a) \leq \frac{1}{a^2} \text{var}(X).$$

Theorem 2.3.29 (Cauchy-Schwarz inequality). For any random variables X and Y for which $E(XY)$ is defined, we have

$$E(XY) \leq \sqrt{E(X^2)E(Y^2)}.$$

Distribution Name	Paramaters	Notation	Range	pmf ($p_X(x)$)	Mean ($\mathbb{E}[X]$)	Variance ($\text{Var}(X)$)
Binomial	$n \in \mathbb{N}, p \in [0, 1]$	$X \sim \text{Bin}(n, p)$	$\{0, 1, \dots, n\}$	$\binom{n}{x} p^x (1-p)^{n-x}$	np	$np(1-p)$
Poisson	$\lambda > 0$	$X \sim \text{Pois}(\lambda)$	$\mathbb{N}_0 = \{0, 1, 2, \dots\}$	$\frac{\lambda^x e^{-\lambda}}{x!}$	λ	λ
Geometric	$p \in (0, 1]$	$X \sim \text{Geo}(p)$	$\mathbb{N} = \{1, 2, 3, \dots\}$	$(1-p)^{x-1} p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Negative Binomial	$p \in (0, 1], r \in \mathbb{N}$	$X \sim \text{NB}(r, p)$	$\{r, r+1, r+2, \dots\}$	$\binom{x-1}{r-1} (1-p)^{x-r} p^r$	$\frac{r}{p}$	$r \frac{1-p}{p^2}$

Table 1: Probability Distributions and Their Properties

Here by Range we mean $\{x \in \mathbb{R} : \exists \omega \in \Omega \text{ where } X(\omega) = x\}$.

Theorem 4.1.1 (Weak law of large numbers). Let $X_1, X_2, \dots, X_n$ be independent random variables with the same distribution. Let $E(X_1) = \mu$ and $\text{var}(X_1) = \sigma^2$ both be finite. Denote the sum $X_1 + \dots + X_n$ by S_n . Then, for any $\epsilon > 0$ we have

$$P\left(\left|\frac{S_n}{n} - \mu\right| > \epsilon\right) \leq \frac{\sigma^2}{n\epsilon^2},$$

which tends to zero when n tends to infinity.

Theorem 4.1.4 (General weak law of large numbers). *Let $X_1, X_2, \dots, X_n$ be independent and identically distributed random variables, with $E(|X_1|) < \infty$ and write $E(X_1) = \mu$. Then for any $\epsilon > 0$ we have*

$$P\left(\left|\frac{S_n}{n} - \mu\right| > \epsilon\right) \rightarrow 0,$$

as $n \rightarrow \infty$.

The following surprising trick is quite standard in probability theory. For $\lambda > 0$, we find that

$$\begin{aligned} \sum_{k > n(\frac{1}{2} + \epsilon)} \binom{n}{k} 2^{-n} &\leq \sum_{k > n(\frac{1}{2} + \epsilon)} e^{\lambda(k - n(\frac{1}{2} + \epsilon))} \binom{n}{k} 2^{-n} \\ &= e^{-\lambda n \epsilon} \sum_{k > n(\frac{1}{2} + \epsilon)} \binom{n}{k} \left(\frac{1}{2} e^{\lambda/2}\right)^k \left(\frac{1}{2} e^{-\lambda/2}\right)^{n-k} \\ &\leq e^{-\lambda n \epsilon} \sum_{k=0}^n \binom{n}{k} \left(\frac{1}{2} e^{\lambda/2}\right)^k \left(\frac{1}{2} e^{-\lambda/2}\right)^{n-k} \\ &= e^{-\lambda n \epsilon} \left(\frac{1}{2} e^{\lambda/2} + \frac{1}{2} e^{-\lambda/2}\right)^n, \end{aligned}$$

1.3 Properties of Probability Measures

In this section we collect and prove a number of useful properties of probability measures. Throughout the section, the sample space is denoted by Ω and $A, B, \dots$ are events in Ω .

Lemma 1.3.1. (a) *For events $A_1, A_2, \dots$ which are pairwise disjoint, we have*

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

(b) $P(A^c) = 1 - P(A)$.

(c) *If $A \subseteq B$, then $P(A) \leq P(B)$.*

More precisely, we have that $P(B) = P(A) + P(B \setminus A)$.

(d) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Let $\Omega = \{\{\omega_1, \omega_2\} : \omega_1, \omega_2 \in S\}$ where $S = S_r \cup S_b$, $S_r = \{r_1, r_2, \dots, r_{2n}\}$, $S_b = \{b_1, b_2, \dots, b_n\}$. An element of Ω represents the two balls drawn, with the $2n$ red balls being labeled $r_1, \dots, r_{2n}$ and similarly for the blue balls. We assume equally likely outcomes, so that $p(\omega) = \frac{1}{|\Omega|} = \frac{1}{\binom{3n}{2}}$ for all $\omega \in \Omega$.

$A_n = \{\omega \in \Omega : \omega_1, \omega_2 \in S_r \text{ or } \omega_1, \omega_2 \in S_b\}$. We write $A_n = A_n^r \cup A_n^b$, where $A_n^r = \{\omega \in \Omega : \omega_1, \omega_2 \in S_r\}$ and $A_n^b = \{\omega \in \Omega : \omega_1, \omega_2 \in S_b\}$. Since A_n^r and A_n^b are disjoint, $\mathbb{P}(A_n) = \mathbb{P}(A_n^r) + \mathbb{P}(A_n^b) = \frac{|A_n^r|}{|\Omega|} + \frac{|A_n^b|}{|\Omega|} = \frac{\binom{2n}{2}}{\binom{3n}{2}} + \frac{\binom{n}{2}}{\binom{3n}{2}} = \frac{(2n)!(3n-2)!}{(2n-2)!(3n)!} + \frac{(n)!(3n-2)!}{(n-2)!(3n)!} = \frac{2(2n-1)}{3(3n-1)} + \frac{(n-1)}{3(3n-1)} \rightarrow \frac{4}{9} + \frac{1}{9} = \frac{5}{9}$ as $n \rightarrow \infty$.